THE VANISHING AND SURJECTIVITY STATEMENTS HOLD FOR THE EXTENSION BY ZERO OF A CONSTANT SHEAF ALONG AN OPEN OF A REDUCED CURVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the normalization and the reduction to the smooth case, the pushforward along the finite morphism, the agreement on the dense open, and the two applications of the restriction-to-open lemma.

Editorial note. The label lemma-somewhat-easier names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing situation and to the enumerated statements by their labels; the eight cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The case of a smooth curve. Same chapter, the companion lemma proving the same statements under the additional assumption that X is smooth. Supplies the result for the normalization of the curve.
- Finiteness and smoothness of the normalization of a curve. Varieties, the lemmas stating that the normalization morphism of a curve is finite and that the normalization of a reduced curve is smooth. Supplies the finite morphism nu : X^nu -> X and the smoothness of X^nu.
- Pushforward of the statements along a finite morphism; restriction to a dense open. Same chapter, the lemma transporting the statements along a finite morphism, and the lemma comparing the statements for a sheaf and for its extension by zero from a dense open. Supplies the transport to X and the final comparison of the two sheaves.

Source reference. Original reference 06AP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15365–15365. Prerequisite pointer invoked at proof line(s) 15570; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A53: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15335–15335. Prerequisite pointer invoked at proof line(s) 15570; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03SG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15548–15548. Prerequisite pointer invoked at proof line(s) 15575; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A5D: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15471–15471. Prerequisite pointer invoked at proof line(s) 15584; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RX: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15491–15491. Prerequisite pointer invoked at proof line(s) 15599; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A52: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15324–15324. Prerequisite pointer invoked at proof line(s) 15567; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BXR: sources/stacks-chow/tags.txt, source lines 14989–14989. Prerequisite pointer invoked at proof line(s) 15578; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B8Y: sources/stacks-chow/tags.txt, source lines 14156–14156. Prerequisite pointer invoked at proof line(s) 15580; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0A52 assume X reduced. Let $j : U \to X$ an open immersion. Let ℓ be a prime number and $\mathcal{F} = j_!\underline{\mathbf{Z}/\ell\mathbf{Z}}$. Then statements (0A53) – (06AP) hold for $\mathcal{F}$.*

Proof. The difference with Lemma 03SG is that here we do not assume X is smooth. Let $\nu : X^\nu \to X$ be the normalization morphism. Then ν is finite (Varieties, Lemma 0BXR) and X^ν is smooth (Varieties, Lemma 0B8Y). Let $j^\nu : U^\nu \to X^\nu$ be the inverse image of U. By Lemma 03SG the result holds for $j^\nu_!\underline{\mathbf{Z}/\ell\mathbf{Z}}$. By Lemma 0A5D the result holds for $\nu_* j^\nu_!\underline{\mathbf{Z}/\ell\mathbf{Z}}$. In general it won't be true that $\nu_* j^\nu_!\underline{\mathbf{Z}/\ell\mathbf{Z}}$ is equal to $j_!\underline{\mathbf{Z}/\ell\mathbf{Z}}$ but we can work around this as follows. As X is reduced the morphism $\nu : X^\nu \to X$ is an isomorphism over a dense open $j' : X' \to X$ (Varieties, Lemma 0BXR). Over this open we have agreement

$$(j')^{-1}(\nu_* j^\nu_!\underline{\mathbf{Z}/\ell\mathbf{Z}}) = (j')^{-1}(j_!\underline{\mathbf{Z}/\ell\mathbf{Z}})$$

Using Lemma 02RX twice for $j' : X' \to X$ and the sheaves above we conclude. □